Kia Motors Sales & Business Analytics

Questions

Customer Analysis

#1. Find the total number of customers.

```
SELECT COUNT(*) AS total_customers
FROM Customers;
```

#2. Find the number of customers from each city.

```
SELECT city,
    COUNT(customer_id) AS total_customers
FROM Customers
GROUP BY city;
```

#3. Find the top 10 customers based on total purchase amount.

```
SELECT c.customer_name,
    SUM(s.sale_amount) AS total_purchase_amount
FROM Customers c
JOIN Sales s
ON c.customer_id = s.customer_id
WHERE s.sale_status = 'Completed'
GROUP BY c.customer_id, c.customer_name
ORDER BY total_purchase_amount DESC
LIMIT 10;
```

#4. Find customers who have purchased more than one vehicle.

```
SELECT c.customer_name,
    COUNT(s.vehicle_id) AS total_vehicles
FROM Customers c
JOIN Sales s
ON c.customer_id = s.customer_id
WHERE s.sale_status = 'Completed'
GROUP BY c.customer_id, c.customer_name
HAVING COUNT(s.vehicle_id) > 1;
```

#5. Find customers who have never purchased a vehicle.

```
SELECT c.customer_name
FROM Customers c
LEFT JOIN Sales s
ON c.customer_id = s.customer_id
AND s.sale_status = 'Completed'
WHERE s.sale_id IS NULL;
```

#6. Find the average purchase amount for each customer.

```
SELECT c.customer_name,
    AVG(s.sale_amount) AS average_purchase_amount
FROM Customers c
JOIN Sales s
ON c.customer_id = s.customer_id
WHERE s.sale_status = 'Completed'
GROUP BY c.customer_id, c.customer_name;
```

#7. Find the customer who spent the highest amount.

```
SELECT c.customer_name,
    SUM(s.sale_amount) AS total_spending
FROM Customers c
JOIN Sales s
ON c.customer_id = s.customer_id
WHERE s.sale_status = 'Completed'
GROUP BY c.customer_id, c.customer_name
ORDER BY total_spending DESC
LIMIT 1;
```

#8. Find customers who purchased a vehicle in 2026.

```
SELECT DISTINCT c.customer_name
FROM Customers c
JOIN Sales s
ON c.customer_id = s.customer_id
WHERE YEAR(s.sale_date) = 2026
AND s.sale_status = 'Completed';
```

#9. Find customers whose purchase amount is greater than the average purchase amount.

```
SELECT c.customer_name,
    s.sale_amount AS purchase_amount
FROM Customers c
JOIN Sales s
ON c.customer_id = s.customer_id
```

```
WHERE s.sale_status = 'Completed'
AND s.sale_amount > (
   SELECT AVG(sale_amount)
   FROM Sales
   WHERE sale_status = 'Completed'
);
```

#10. Find the number of customers in each state.

```
SELECT state,
      COUNT(customer_id) AS total_customers
FROM Customers
GROUP BY state;
```

Vehicle Analysis

#11. Find the total number of vehicle models.

```
SELECT COUNT(*) AS total_vehicle_models
FROM Vehicle_Models;
```

#12. Find the number of vehicles available for each model.

```
SELECT vm.model_name,
      COUNT(v.vehicle_id) AS total_vehicles
FROM Vehicle_Models vm
LEFT JOIN Vehicles v
ON vm.model_id = v.model_id
AND v.availability_status = 'Available'
GROUP BY vm.model_id, vm.model_name;
```

#13. Find the top 10 vehicle models based on sales.

```
SELECT vm.model_name,
    SUM(sd.quantity) AS total_sales
FROM Vehicle_Models vm
JOIN Sales_Details sd
ON vm.model_id = sd.model_id
JOIN Sales s
ON sd.sale_id = s.sale_id
WHERE s.sale_status = 'Completed'
GROUP BY vm.model_id, vm.model_name
ORDER BY total_sales DESC
LIMIT 10;
```

#14. Find the most expensive vehicle model.

```
SELECT model_name,
    base_price
FROM Vehicle_Models
ORDER BY base_price DESC
LIMIT 1;
```

#15. Find the least expensive vehicle model.

```
SELECT model_name,
    base_price
FROM Vehicle_Models
ORDER BY base_price ASC
LIMIT 1;
```

#16. Find vehicles that have never been sold.

```
SELECT v.registration_number,
    vm.model_name
FROM Vehicles v
JOIN Vehicle_Models vm
ON v.model_id = vm.model_id
LEFT JOIN Sales s
ON v.vehicle_id = s.vehicle_id
AND s.sale_status = 'Completed'
WHERE s.sale_id IS NULL;
```

#17. Find the average selling price of each vehicle model.

```
SELECT vm.model_name,
    AVG(sd.selling_price) AS average_selling_price
FROM Vehicle_Models vm
JOIN Sales_Details sd
ON vm.model_id = sd.model_id
JOIN Sales s
ON sd.sale_id = s.sale_id
WHERE s.sale_status = 'Completed'
GROUP BY vm.model_id, vm.model_name;
```

#18. Find the vehicle model generating the highest revenue.

```
SELECT vm.model_name,
    SUM(sd.selling_price * sd.quantity) AS total_revenue
```

```
FROM Vehicle_Models vm
JOIN Sales_Details sd
ON vm.model_id = sd.model_id
JOIN Sales s
ON sd.sale_id = s.sale_id
WHERE s.sale_status = 'Completed'
GROUP BY vm.model_id, vm.model_name
ORDER BY total_revenue DESC
LIMIT 1;
```

#19. Find the number of vehicles sold in each fuel type.

```
SELECT vm.fuel_type,
      SUM(sd.quantity) AS vehicles_sold
FROM Vehicle_Models vm
JOIN Sales_Details sd
ON vm.model_id = sd.model_id
JOIN Sales s
ON sd.sale_id = s.sale_id
WHERE s.sale_status = 'Completed'
GROUP BY vm.fuel_type;
```

#20. Find the most popular vehicle model in each city.

```
SELECT city, model_name, total_sales
FROM (
   SELECT d.city,
         vm.model_name,
         SUM(sd.quantity) AS total_sales,
```

```
        RANK() OVER (
            PARTITION BY d.city
            ORDER BY SUM(sd.quantity) DESC
        ) AS ranking
    FROM Dealerships d
    JOIN Sales s
    ON d.dealership_id = s.dealership_id
    JOIN Vehicles v
    ON s.vehicle_id = v.vehicle_id
    JOIN Vehicle_Models vm
    ON v.model_id = vm.model_id
    JOIN Sales_Details sd
    ON s.sale_id = sd.sale_id
    WHERE s.sale_status = 'Completed'
    GROUP BY d.city, vm.model_id, vm.model_name
) AS x
WHERE ranking = 1;
```

Dealership Analysis

#21. Find the total number of dealerships.

```
SELECT COUNT(*) AS total_dealerships
FROM Dealerships;
```

#22. Find the number of dealerships in each city.

```
SELECT city,
    COUNT(dealership_id) AS total_dealerships
FROM Dealerships
GROUP BY city;
```

#23. Find the top 10 dealerships based on total sales revenue.

```
SELECT d.dealership_name,
    SUM(s.sale_amount) AS total_revenue
FROM Dealerships d
JOIN Sales s
ON d.dealership_id = s.dealership_id
WHERE s.sale_status = 'Completed'
GROUP BY d.dealership_id, d.dealership_name
ORDER BY total_revenue DESC
LIMIT 10;
```

#24. Find the dealership with the highest number of vehicles sold.

```
SELECT d.dealership_name,
    SUM(sd.quantity) AS vehicles_sold
FROM Dealerships d
JOIN Sales s
ON d.dealership_id = s.dealership_id
JOIN Sales_Details sd
ON s.sale_id = sd.sale_id
WHERE s.sale_status = 'Completed'
GROUP BY d.dealership_id, d.dealership_name
ORDER BY vehicles_sold DESC
LIMIT 1;
```

#25. Find dealerships that have not sold any vehicle.

```
SELECT d.dealership_name
FROM Dealerships d
LEFT JOIN Sales s
ON d.dealership_id = s.dealership_id
AND s.sale_status = 'Completed'
WHERE s.sale_id IS NULL;
```

#26. Find the average sales value for each dealership.

```
SELECT d.dealership_name,
    AVG(s.sale_amount) AS average_sales_value
FROM Dealerships d
JOIN Sales s
ON d.dealership_id = s.dealership_id
```

```
WHERE s.sale_status = 'Completed'
GROUP BY d.dealership_id, d.dealership_name;
```

#27. Find dealerships whose revenue is higher than the average dealership revenue.

```
SELECT d.dealership_name,
      SUM(s.sale_amount) AS total_revenue
FROM Dealerships d
JOIN Sales s
ON d.dealership_id = s.dealership_id
WHERE s.sale_status = 'Completed'
GROUP BY d.dealership_id, d.dealership_name
HAVING SUM(s.sale_amount) > (
   SELECT AVG(total_revenue)
   FROM (
      SELECT SUM(s2.sale_amount) AS total_revenue
      FROM Sales s2
      WHERE s2.sale_status = 'Completed'
      GROUP BY s2.dealership_id
   ) AS dealership_revenue
);
```

#28. Find the dealership generating the lowest revenue.

```
SELECT d.dealership_name,
      COALESCE(SUM(s.sale_amount), 0) AS total_revenue
FROM Dealerships d
LEFT JOIN Sales s
ON d.dealership_id = s.dealership_id
```

```
AND s.sale_status = 'Completed'
GROUP BY d.dealership_id, d.dealership_name
ORDER BY total_revenue ASC
LIMIT 1;
```

Sales & Revenue Analysis

#29. Calculate the total number of vehicles sold.

```
SELECT SUM(sd.quantity) AS total_vehicles_sold
FROM Sales s
JOIN Sales_Details sd
ON s.sale_id = sd.sale_id
WHERE s.sale_status = 'Completed';
```

#30. Calculate the total sales revenue.

```
SELECT SUM(sale_amount) AS total_sales_revenue
FROM Sales
WHERE sale_status = 'Completed';
```

#31. Calculate the average vehicle selling price.

```
SELECT AVG(sd.selling_price) AS average_vehicle_selling_price
FROM Sales s
JOIN Sales_Details sd
ON s.sale_id = sd.sale_id
WHERE s.sale_status = 'Completed';
```

#32. Find monthly sales revenue for 2026.

```
SELECT MONTH(sale_date) AS month,
    SUM(sale_amount) AS monthly_revenue
FROM Sales
WHERE YEAR(sale_date) = 2026
AND sale_status = 'Completed'
GROUP BY MONTH(sale_date)
ORDER BY month;
```

#33. Find the month with the highest sales revenue.

```
SELECT MONTH(sale_date) AS month,
    SUM(sale_amount) AS total_revenue
FROM Sales
WHERE YEAR(sale_date) = 2026
AND sale_status = 'Completed'
GROUP BY MONTH(sale_date)
ORDER BY total_revenue DESC
LIMIT 1;
```

#34. Find the month with the lowest sales revenue.

```
SELECT MONTH(sale_date) AS month,
    SUM(sale_amount) AS total_revenue
FROM Sales
WHERE YEAR(sale_date) = 2026
AND sale_status = 'Completed'
GROUP BY MONTH(sale_date)
ORDER BY total_revenue ASC
LIMIT 1;
```

#35. Find the number of vehicles sold for each fuel type.

```
SELECT vm.fuel_type,
      SUM(sd.quantity) AS vehicles_sold
FROM Vehicle_Models vm
JOIN Sales_Details sd
ON vm.model_id = sd.model_id
JOIN Sales s
ON sd.sale_id = s.sale_id
WHERE s.sale_status = 'Completed'
GROUP BY vm.fuel_type;
```

#36. Calculate the percentage of cancelled sales.

```
SELECT
   (SUM(CASE
      WHEN sale_status = 'Cancelled' THEN 1
      ELSE 0
   END) / COUNT(*)) * 100 AS cancelled_percentage
FROM Sales;
```

#37. Find the city generating the highest sales revenue.

```
SELECT d.city,
      SUM(s.sale_amount) AS total_revenue
FROM Dealerships d
JOIN Sales s
ON d.dealership_id = s.dealership_id
```

```
WHERE s.sale_status = 'Completed'
GROUP BY d.city
ORDER BY total_revenue DESC
LIMIT 1;
```

#38. Find the daily sales revenue for a selected month.

```
SELECT sale_date,
    SUM(sale_amount) AS daily_revenue
FROM Sales
WHERE YEAR(sale_date) = 2026
AND MONTH(sale_date) = 8
AND sale_status = 'Completed'
GROUP BY sale_date
ORDER BY sale_date;
```

Employee Analysis

#39. Find the total number of sales employees.

```
SELECT COUNT(*) AS total_sales_employees
FROM Employees
WHERE job_role LIKE '%Sales%';
```

#40. Find the number of employees working at each dealership.

```
SELECT d.dealership_name,
    COUNT(e.employee_id) AS total_employees
FROM Dealerships d
JOIN Employees e
```

```
ON d.dealership_id = e.dealership_id
GROUP BY d.dealership_id, d.dealership_name;
```

#41. Find employees who have generated more than the average sales revenue.

```
SELECT e.employee_name,
      SUM(s.sale_amount) AS total_revenue
FROM Employees e
JOIN Sales s
ON e.employee_id = s.employee_id
WHERE s.sale_status = 'Completed'
GROUP BY e.employee_id, e.employee_name
HAVING SUM(s.sale_amount) > (
   SELECT AVG(total_revenue)
   FROM (
      SELECT SUM(s2.sale_amount) AS total_revenue
      FROM Sales s2
      WHERE s2.sale_status = 'Completed'
      GROUP BY s2.employee_id
   ) AS employee_revenue
);
```

#42. Find the employee with the highest number of vehicle sales.

```
SELECT e.employee_name,
      SUM(sd.quantity) AS vehicles_sold
FROM Employees e
JOIN Sales s
ON e.employee_id = s.employee_id
```

```
JOIN Sales_Details sd
ON s.sale_id = sd.sale_id
WHERE s.sale_status = 'Completed'
GROUP BY e.employee_id, e.employee_name
ORDER BY vehicles_sold DESC
LIMIT 1;
```

#43. Find the employee who generated the highest revenue.

```
SELECT e.employee_name,
    SUM(s.sale_amount) AS total_revenue
FROM Employees e
JOIN Sales s
ON e.employee_id = s.employee_id
WHERE s.sale_status = 'Completed'
GROUP BY e.employee_id, e.employee_name
ORDER BY total_revenue DESC
LIMIT 1;
```

Payment Analysis

#44. Find the most frequently used payment method.

```
SELECT payment_method,
    COUNT(*) AS total_transactions
FROM Payments
GROUP BY payment_method
ORDER BY total_transactions DESC
LIMIT 1;
```

#45. Calculate total revenue received through each payment method.

```
SELECT payment_method,
    SUM(amount) AS total_revenue
FROM Payments
WHERE payment_status = 'Successful'
GROUP BY payment_method
ORDER BY total_revenue DESC;
```

#46. Find the number of successful and failed payments.

```
SELECT payment_status,
    COUNT(*) AS total_payments
FROM Payments
GROUP BY payment_status;
```

#47. Find the payment method with the highest failure rate.

```
SELECT
   payment_method,
   (SUM(CASE
      WHEN payment_status = 'Failed' THEN 1
      ELSE 0
   END) / COUNT(*)) * 100 AS failure_rate
FROM
   Payments
GROUP BY payment_method
ORDER BY failure_rate DESC
LIMIT 1;
```

Service Analysis

#48. Find the number of service records for each vehicle model.

```
SELECT vm.model_name,
      COUNT(sr.service_id) AS total_service_records
FROM Vehicle_Models vm
LEFT JOIN Vehicles v
ON vm.model_id = v.model_id
LEFT JOIN Service_Records sr
ON v.vehicle_id = sr.vehicle_id
GROUP BY vm.model_id, vm.model_name;
```

#49. Calculate the average service cost for each vehicle model.

```
SELECT vm.model_name,
      AVG(sr.service_cost) AS average_service_cost
```

```
FROM Vehicle_Models vm
JOIN Vehicles v
ON vm.model_id = v.model_id
JOIN Service_Records sr
ON v.vehicle_id = sr.vehicle_id
GROUP BY vm.model_id, vm.model_name;
```

#50. Find vehicles whose service cost is higher than the average service cost.

```
SELECT v.registration_number,
       vm.model_name,
       sr.service_cost
FROM Vehicles v
JOIN Vehicle_Models vm
ON v.model_id = vm.model_id
JOIN Service_Records sr
ON v.vehicle_id = sr.vehicle_id
WHERE sr.service_cost > (
   SELECT AVG(service_cost)
   FROM Service_Records
);
```